

L01 Exploring Real-World Applications of Computer Vision

Technical Analysis: Facial Recognition for Security

1. Description of the Application

Facial recognition is a technology that identifies or verifies a person by looking at their face. In security, its main purpose is to replace physical keys, plastic ID cards, and text passwords. Today, people use it every day to unlock their mobile phones, log into bank accounts, and enter secure office buildings. Police also use it to find missing people or suspects in crowded areas.

2. Technology Behind It

Facial recognition turns human facial features into computer numbers to confirm who you are.

- **Main System Components:**

- **Cameras:** High-quality cameras capture photos or videos of your face. Some systems use special infrared cameras that can see your face in the dark.
- **Database:** A secure computer storage system that holds pictures or digital codes of allowed users.

- **The Role of Computer Vision:**

- **Face Detection:** The computer looks at a picture and finds exactly where the human face is. It ignores the background.
- **Finding Landmarks:** The system measures specific distances on your face. For example, it measures the distance between your eyes, the width of your nose, and the shape of your chin.
- **Matching:** Artificial Intelligence (AI) turns these measurements into a unique math code, like a fingerprint for your face. The computer checks this code against its database to see if you are allowed in.

3. Benefits and Challenges

Advantages

- **Great Convenience:** You can never lose or forget your face, unlike a password or a physical key.

- **Touchless Security:** You do not need to touch any dirty buttons or scanners to open a door. You just look at the camera.

Challenges and Limitations

- **Cheating the System (Spoofing):** Sometimes, simple systems can be tricked if a person holds up a high-quality printed photo or a video of the real owner.
- **The Big Security Risk:** If a hacker steals your password, you can change it. But if a hacker steals your face data from a computer server, you cannot change your face. Your digital identity is stolen forever.
- **No Privacy:** If cities put these cameras everywhere, people feel like they are being watched 24/7. This takes away personal freedom.

4. Reflection

In the future, engineers will focus on protecting our data. They are building systems that can check your face without saving your real photo on the internet. This will keep our identities much safer.

To conclude, this technology affects society in two ways. On the positive side, it makes our smartphones secure and helps airports check passengers faster. On the negative side, if governments or companies use it without strict rules, it can destroy personal privacy and create a world where no one can be anonymous.